3.1.4 Aggregate Production

The overall objective in aggregate processing is to economically produce aggregates that are appropriate for the end use. Particularly for the structural and functional needs of the more demanding asphalt paving mixtures, aggregates must have particular particle shape, surface texture, and grading or particle-size distribution characteristics in addition to the inherent strength and durability characteristics.

Extraction

The first stage in the production of aggregates is the extraction or mining of the source material from its deposited condition. The material must be accessed through an open or "working" face of the deposit. Many times, overburden (unwanted or unusable material) that lies above the aggregate deposit must be removed or stripped before the working face can be established. In cases where overburden is primarily soil, removal is typically accomplished with common earth-moving equipment such as scrapers. Removal of overburden containing more highly consolidated soil and rock fragments may require bulldozers, front-end loaders, and haul trucks.

In their natural state in a dry pit, gravel and sand are typically in a relatively unconsolidated condition. For this condition, mining of the working face is usually accomplished with front-end loaders or power shovels. For pits with larger and taller working faces, a bulldozer is often employed to push the material down the face to then allow for the use of loaders or shovels. The extracted material is then loaded into haul trucks or placed on a conveyor for transportation to the processing plant for screening, crushing, and washing.

Some deposits of sand and gravel lie beneath bodies of water, such as rivers and lakes. Extraction of these materials usually requires dredges, draglines, or large suction pumps.

Quarry stone is commonly extracted by drilling and blasting the rock formation. Mechanical extraction tools, such as rock hammers for further rock fragmentation after blasting, are sometimes necessary. Equipment and methodology for handling and transporting the rock fragments or shot rock is essentially the same as that for gravel and sand.

Crushing

Most HMA mixtures use a maximum aggregate size of 1½ inches (37.5 millimeters) in diameter or less. In addition, they require some minimum level of aggregate surface roughness and shape for developing the necessary aggregate structure to resist pavement loads and deformation. Hence, crushing of the aggregate is usually necessary to further reduce gravel and rock fragments to proper particle sizes and to create appropriate shape and surface characteristics.

The two main types of crushers are called compression and impact—so named according to the manner in which force is applied to the fracture

mechanisms occurring in the rock. Compression-based equipment, such as jaw, cone, gyratory, and roll crushers, essentially squeeze or compress rock between their crushing surfaces and fracture it. In impact crushers, rock particles are impacted against other rock particles and against surfaces inside the crusher at high speeds. The rock particles shatter in impact crushers.

The amount of reduction in particle size, the shape of the aggregate particles, and the amount of fines or minus No. 4 (4.75 millimeters) material vary with the different sizes and kinds of crushers, as well as with the rock type and rate at which the rock is fed into the crusher. In order to yield the most appropriate end products, multiple crushing stages with optimized crushing equipment for each particular stage coupled with appropriate screening sequences are common.

Screening

The screening operation is the part of the processing procedure in which aggregates are separated according to size. Material may be passed through single-, double-, or even triple-screening applications in order to make the various needed sizes of aggregate. Before or during screening applications, material can be routed to crushing operations. Additionally, material can be conveyed to washing operations before, during, or after screening applications.

Several types of equipment are used for screening, but the most common is the vibrating inclined screen. This equipment allows the slope, vibration speed, stroke, and direction of rotation of the screen surface to be adjusted as needed. Typically, units with multiple-deck screens are used in production facilities. Screen surfaces are available in various types and materials and with a variety of sizes and shapes of screen openings.

Washing

Aggregates may need to be washed during production to remove dust generated during crushing. Depending on the nature of sand and gravel deposits, washing may also be needed to remove silts, clays, and other deleterious substances such as coal and soft or lightweight fragments.

The various types of washing equipment include high-pressure spray nozzles, screw washers, log washers, and rotary scrubbers. Sprayers mounted above the screens might adequately rinse some unwanted materials, such as dust, from aggregates. Other materials, such as more highly plastic clays and cemented aggregates, may need the rigorous agitation of a log washer.

Handling and Stockpiling

The handling and stockpiling of aggregates is an integral component of aggregate processing for producing a quality product. The methods used for the pickup or loading of material by shovels and loader buckets should avoid segregation. Feed bins, chutes, gates, and conveyors should be sized and operated in such a way that a proper and consistent rate of feed is maintained to crushers, screens, and washers.

Each stage of conveying and storing aggregates must minimize contamination, degradation, and segregation. Degradation can mean not only the breakdown of aggregates into smaller pieces but also the creation of dust. Excessive dropping of aggregates from conveyors, pushing of aggregates with bulldozers, handling of aggregates by front-end loaders, and buildup of stockpiles to excessive heights all contribute to degradation.

Segregation is the separation of larger particles from smaller particles, leaving a nonhomogenous and unrepresentative blend of particle sizes. Because movement and vibration cause segregation, every stage of handling and processing can contribute to it. On conveyor belts and in haul trucks, smaller particles migrate downward. More segregation takes place at the conveyor end and as the material is dumped from the truckbed. In these cases, larger particles flow away from the belt or the truckbed and the finer particles drop more vertically and under the belt or truckbed (see Figure 3.3). The worst segregation occurs in single-cone stockpiles where material is dropped from a tall, stationary conveyor belt to build the stockpile. Segregation can be controlled with better stockpiling techniques, such as using radial stackers. This equipment allows the formation of a continuous layered stockpile. Each layer will have segregation, but it occurs in a more finite volume because load-out of the multiple layers negates the individual-layer segregation. Another way to control or reduce segregation is to screen aggregates into a greater number of fractions with fewer particle sizes in each fraction.

Sampling

Material sampling is a key component of aggregate quality-control, yet so often it is overlooked or taken for granted. It is imperative that samples be obtained in such a manner that they are representative of the aggregate product. Too often, samples are manually taken from the faces of coarse-aggregate stockpiles where, due to the common segregation that occurs in the building of stockpiles, the samples are not representative of the whole pile but are assumed to be. In actuality, they represent only a small volume of aggregate in the particular sample locations.

The methodologies of AASHTO T 2 (ASTM D 75) should be followed when sampling aggregates. Typically, the best sampling locations are wherever the full material flow-stream can be sampled, such as from discharge streams, from bins or conveyor belt ends, or from the full cross section of conveyor-belt surfaces. If sampling must be done from stockpiles, the procedures detailed in the appendix of AASHTO T 2 should be followed.

3.1.5 Aggregate Properties

Concerning aggregates for hot mix asphalt, *The Aggregate Handbook*, published by the National Stone Sand and Gravel Association, states: "The ideal aggregate has proper particle size and grading, is strong and tough, and consists of angular, nearly equidimensional (cubical) particles with

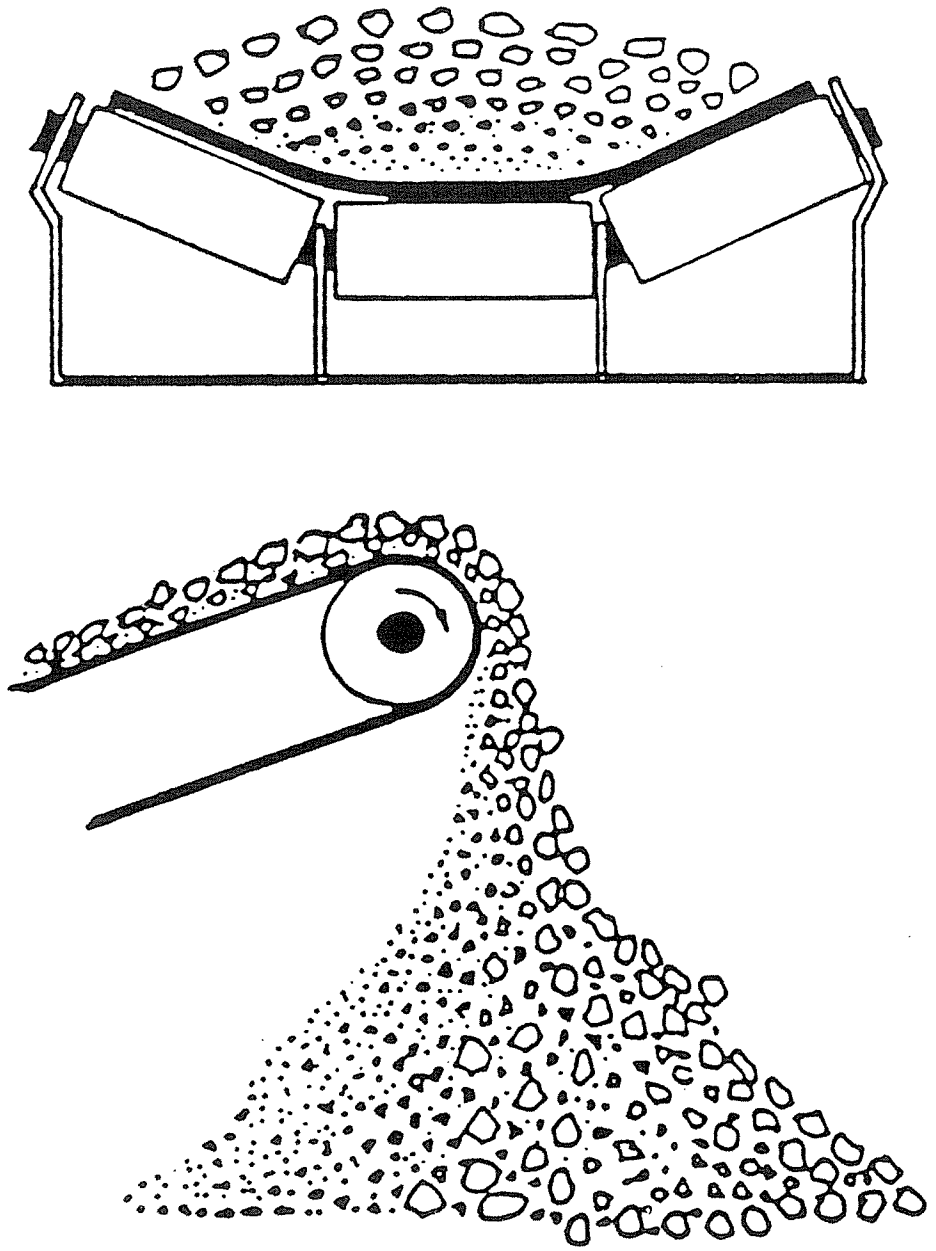


Figure 3.3 Aggregate Segregation

moderately low porosity. The surface of the aggregate is clean, rough, and hydrophobic, which means it has an affinity for asphalt rather than water."

There is no question that aggregates of the highest quality or with the best properties are preferred. However, selecting an aggregate material for use in an asphalt pavement also depends on the availability and cost of the material, as well as the type of construction.

The suitability of aggregates for asphalt construction is most commonly determined by evaluating the material in terms of the following characteristics:

- Maximum particle size and grading
- Particle shape and surface texture
- Toughness
- Pore structure, porosity, and absorption
- Specific gravity
- Freeze-thaw and weathering resistance
- Deleterious substances
- Affinity for water or asphalt
- Cleanliness

Maximum Particle Size and Grading

Two key properties of an aggregate that go hand in hand are maximum particle size and grading. ASTM D 8 defines the maximum size of an aggregate as "the smallest sieve opening through which the entire amount of aggregate is required to pass." AASHTO MP 323 refers to the maximum size of an aggregate for Superpave mixes only as "one size larger than the nominal maximum aggregate size."

AASHTO MP 323 defines it as "one sieve size larger than the first sieve to retain more than 10 percent of the material." Care must be exercised when using and referring to the terms. For good paving results, the lift thickness of HMA should be at least 2 to 2½ times the maximum aggregate size in the mix or 3 to 4 times the nominal maximum aggregate size in the mix.

The grading or gradation of an aggregate is the distribution of the various particle sizes. Gradation is typically determined by both a washed sieve analysis and a dry sieve analysis. ASTM C 136 (AASHTO T 27) details the procedure for dry sieving.

In this procedure, the mass of a sample of dry aggregate is determined and then the sample is passed through a series of sieves with progressively smaller openings (see Figure 3.4). Once separated, the mass of material retained on each sieve is measured and compared with the total sample weight. The aggregate gradation is ultimately expressed as the cumulative percentage passing each sieve size. Figure 3.5 gives an example of sieve analysis data for an aggregate that contains both fine and coarse particles. A graphical plot of the particle distribution is also shown.

Dry sieving alone may be satisfactory for general checks during aggregate production but does not appropriately determine the amount of fine particles smaller than the No. 200 (75 microns) sieve. Therefore, before performing the dry sieve analysis, a washed sieve analysis should be made as detailed in AASHTO T 11 (ASTM C 117). They describe the procedure in this way: "A sample of the aggregate is washed in a prescribed manner, using either plain water or water containing a wetting agent, as specified. The decanted wash water, containing suspended and dissolved material, is passed through a 75 microns (No. 200) sieve. The loss in mass resulting from the wash treatment is calculated as mass percent of the original sample. . . ."

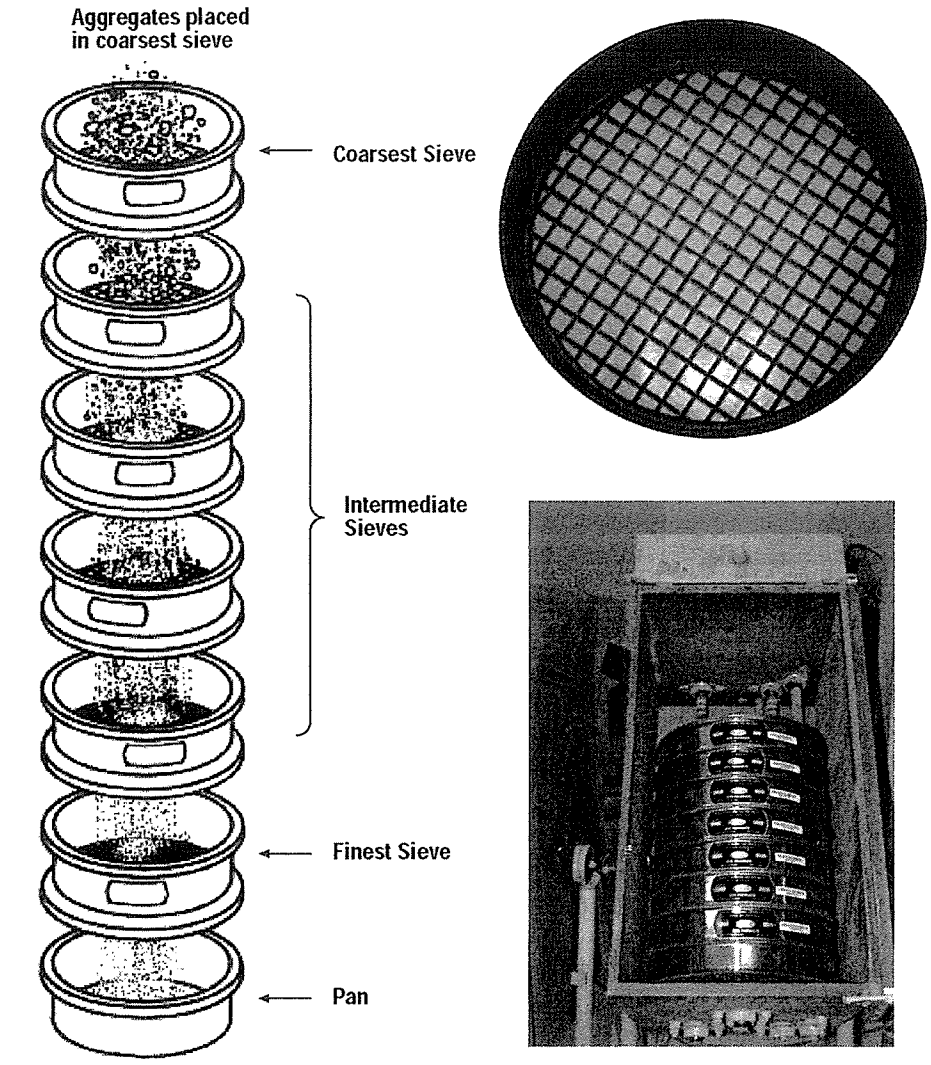


Figure 3.4 Sieving of Aggregate

Table 3.3 shows the various standard sizes and grading bands of coarse aggregate as contained in AASHTO M 43 (ASTM D 448).

Particle Shape and Surface Texture

Particle shape refers to the form and contour of the individual aggregate particles (angular versus round, cubic versus flat or elongated). Surface texture refers to the extent of roughness of the surface of the aggregate particles.

Although two separate characteristics—particle shape and surface texture of aggregates—are often considered collectively in their roles in HMA workability, compaction, and deformation resistance (see Figure 3.6), both fine- and coarse-size fractions of crushed stone and gravel have irregular or angular shapes as well as rougher surface textures. Natural (uncrushed) gravels and sands have generally rounded shapes and smoother textures.

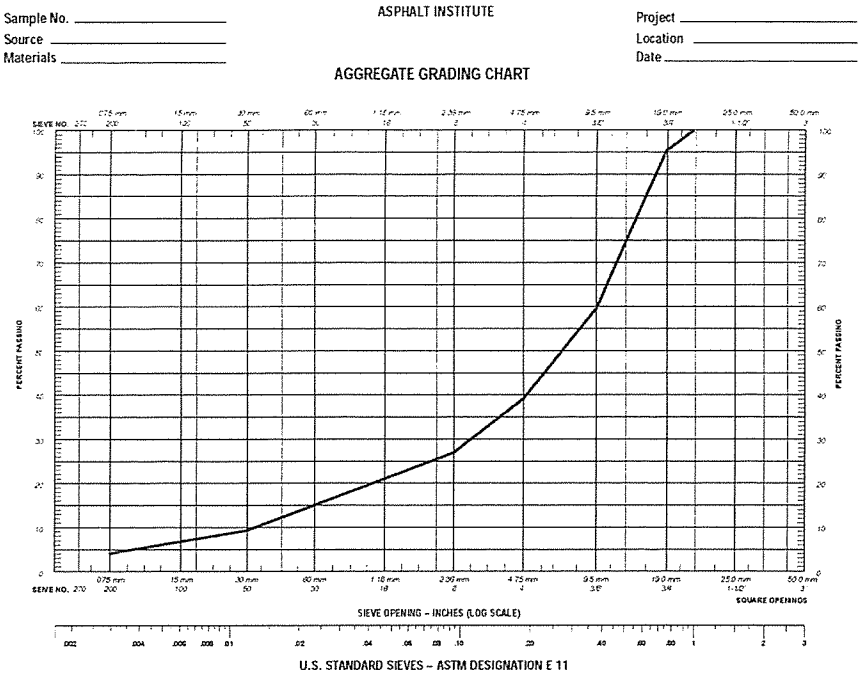


Figure 3.5 Example of Aggregate Sieve Analysis



Figure 3.6 Round Versus Fractured Particles

Particle shape changes the workability and strength characteristics of the paving mix as well as the compaction needed to obtain the required density. Irregular or angular particles, such as crushed stone and gravels, and some natural gravels and sands, tend to interlock when compacted and resist displacement. Best interlock is generally obtained with sharp-cornered, cubical-shaped particles; interlock is generally least effective with round particles. Flat or elongated particles tend to hinder compaction and to

Table 3.3
Coarse-Aggregate Standard Grading Sizes

Size No.	Nominal Size, Square Openings	Amounts Finer than Each Laboratory Sieve (Square Openings), weight percent														
		4 in. (100 mm)	3 ½ in. (90 mm)	3 in. (75 mm)	2 ½ in. (63 mm)	2 in. (50 mm)	1 ½ in. (37.5 mm)	1 in. (25 mm)	¾ in. (19 mm)	½ in. (12.5 mm)	⅜ in. (9.5 mm)	No. 4 (4.75 mm)	No. 8 (2.36 mm)	No. 16 (1.18 mm)	No. 50 (300 μm)	No. 100 (150 μm)
1	3½ to 1½ in. (90 to 37.5 mm)	100	90 to 100	–	25 to 60	–	0 to 15	–	0 to 5	–	–	–	–	–	–	–
2	2½ to 1½ in. (63 to 37.5 mm)	–	–	100	90 to 100	35 to 70	0 to 15	–	0 to 5	–	–	–	–	–	–	–
24	2½ to ¾ in. (63 to 19 mm)	–	–	100	100	–	25 to 60	–	0 to 10	0 to 5	–	–	–	–	–	–
3	2 to 1 in. (50 to 25 mm)	–	–	–	100	90 to 100	35 to 70	0 to 15	–	0 to 5	–	–	–	–	–	–
357	2 in. to No. 4 (50 to 4.75 mm)	–	–	–	–	95 to 100	–	35 to 70	–	10 to 30	–	0 to 5	–	–	–	–
4	1½ in to ¾ in. (37.5 to 19 mm)	–	–	–	–	100	90 to 100	–	20 to 55	0 to 15	–	0 to 5	–	–	–	–
467	1½ to No. 4 (37.5 to 4.75 mm)	–	–	–	–	100	95 to 100	–	35 to 70	–	10 to 30	0 to 5	–	–	–	–
5	1 to ½ in. (25 to 12.5 mm)	–	–	–	–	–	100	90 to 100	20 to 55	0 to 10	0 to 5	–	–	–	–	–
56	1 to ⅜ in. (25 to 9.5 mm)	–	–	–	–	–	100	90 to 100	40 to 70	15 to 35	0 to 15	0 to 5	–	–	–	–
57	1 in. to No. 4 (25 to 4.75 mm)	–	–	–	–	–	100	95 to 100	–	25 to 60	–	0 to 10	0 to 5	–	–	–

Table 3.3
Coarse-Aggregate Standard Grading Sizes (Continued)

6	¾ to ⅜ in. (19 to 9.5 mm)	–	–	–	–	–	–	100	90 to 100	20 to 55	0 to 15	0 to 5	–	–	–	–
67	¾ in. to No. 4 (19 to 4.75 mm)	–	–	–	–	–	–	100	90 to 100	–	20 to 55	0 to 10	0 to 5	–	–	–
68	¾ in. to No. 8 (19 to 2.36 mm)	–	–	–	–	–	–	100	90 to 100	–	30 to 65	5 to 25	0 to 10	0 to 5	–	–
7	½ in. to No. 4 (12.5 to 4.75 mm)	–	–	–	–	–	–	–	100	90 to 100	40 to 70	0 to 15	0 to 5	–	–	–
78	½ in. to No. 8 (12.5 to 2.36 mm)	–	–	–	–	–	–	–	100	90 to 100	40 to 75	5 to 25	0 to 10	0 to 5	–	–
8	⅜ in. to No. 8 (9.5 to 2.36 mm)	–	–	–	–	–	–	–	–	100	85 to 100	10 to 30	0 to 10	0 to 5	–	–
89	⅜ in. to No. 16 (9.5 to 1.18 mm)	–	–	–	–	–	–	–	–	100	90 to 100	20 to 55	5 to 30	0 to 10	0 to 5	–
9	No. 4 to No. 16 (4.75 to 1.18 mm)	–	–	–	–	–	–	–	–	–	100	85 to 100	10 to 40	0 to 10	0 to 5	–
10	No. 4 to 0[a] (4.75 mm)	–	–	–	–	–	–	–	–	–	100	85 to 100	–	–	–	10 to 30

[a] Screenings

break during compaction, leaving aggregate particles with uncoated faces, which in turn decreases strength. Also, flat and elongated particles are more susceptible to segregation.

Like particle shape, the surface texture influences the workability, compactability, and strength of asphalt paving mixtures. A rough, sandpaper-like surface texture (as opposed to a smooth glass-like surface) tends to increase the strength of the paving mix and requires additional asphalt to overcome the loss of workability. Voids in the compacted mineral aggregate are also typically greater, a characteristic that provides additional space for the required increase of asphalt.

Asphalt paving mixes that carry heavy traffic loads and are highly resistant to rutting and deformation usually contain rough-textured, crushed, and angular aggregates, both coarse and fine. These mixes require greater care and effort to place and compact.

Many asphalt mixes in pavements for medium to low traffic contain both angular, rough-textured aggregate particles and smoother, rounded aggregate particles. The coarse aggregate fraction is usually crushed stone or crushed gravel, and the fine aggregate is usually all natural sand (rounded particles) or a combination with stone screenings. Such mixes rely principally on crushed aggregate for strength and rounded sand particles for workability and ease of compaction.

Rounded particles, such as most natural gravels (uncrushed) and sands from streambeds, have been used successfully in light duty asphalt pavements, such as parking lots and walking trails. However, because a higher density is possible with round aggregate particles, the amount of asphalt in the mix is usually a more critical factor in such dense-graded mixes. Several tests are available for measuring aggregate particle shape and surface texture properties. The more common tests include the following:

- Percentage of fractured particles in coarse aggregate (referred to sometimes as coarse aggregate angularity)—ASTM D 5821
- Flat and elongated particles in coarse aggregate—ASTM D 4791
- Uncompacted void content of fine aggregate (fine-aggregate angularity)—ASTM C 1252 or AASHTO T 304
- Uncompacted void content of coarse aggregate—AASHTO T 326

Another test that is sometimes used is the particle index—ASTM D 3398.

Toughness

An aggregate's toughness is measured by its ability to resist wear and degradation. Aggregates are subject to additional crushing and abrasive wear during manufacture, placing, and compaction of asphalt paving mixes and so must exhibit a certain degree of toughness. Aggregates in HMA layers at or near the pavement surface are also subject to abrasion under traffic loads and require more toughness than aggregates in the lower layers where loads have dissipated or are not as concentrated. Relatively high resistance to wear, as indicated by a low percentage of abrasion loss, is a desirable characteristic of aggregates to be used in asphalt pavement surface layers. Aggregates having higher abrasion losses, within

limits, may generally be used in lower pavement layers where they will not be subject to the high stresses caused by traffic.

The Los Angeles Abrasion test has been the primary method to measure the wear or abrasion resistance of mineral aggregate. Equipment and procedures for this test are detailed in AASHTO T 96 (ASTM C 131). The abrasion machine is illustrated in Figure 3.7. The test involves sieving and proportioning the aggregate to a standard grading, and then placing the sample into the drum of the abrasion device along with a specified number of steel balls. The material is then subjected to abrasion and impact forces inside the drum as it rotates through 500 revolutions. The sample is then sieved over a No. 12 (1.70 millimeters) sieve. From the mass of the retained material, the mass of loss material is determined and then calculated as a percentage of loss of the original mass. The common criterion in specifications for dense graded mixes is a maximum allowable abrasion loss of 40 to 50 percent.

AASHTO T 327 (ASTM D 6928) is the Micro-Deval test and is similar to the Los Angeles Abrasion test except that the drum size and sample size are about 25 percent of that for the Los Angeles Abrasion test. Also, the

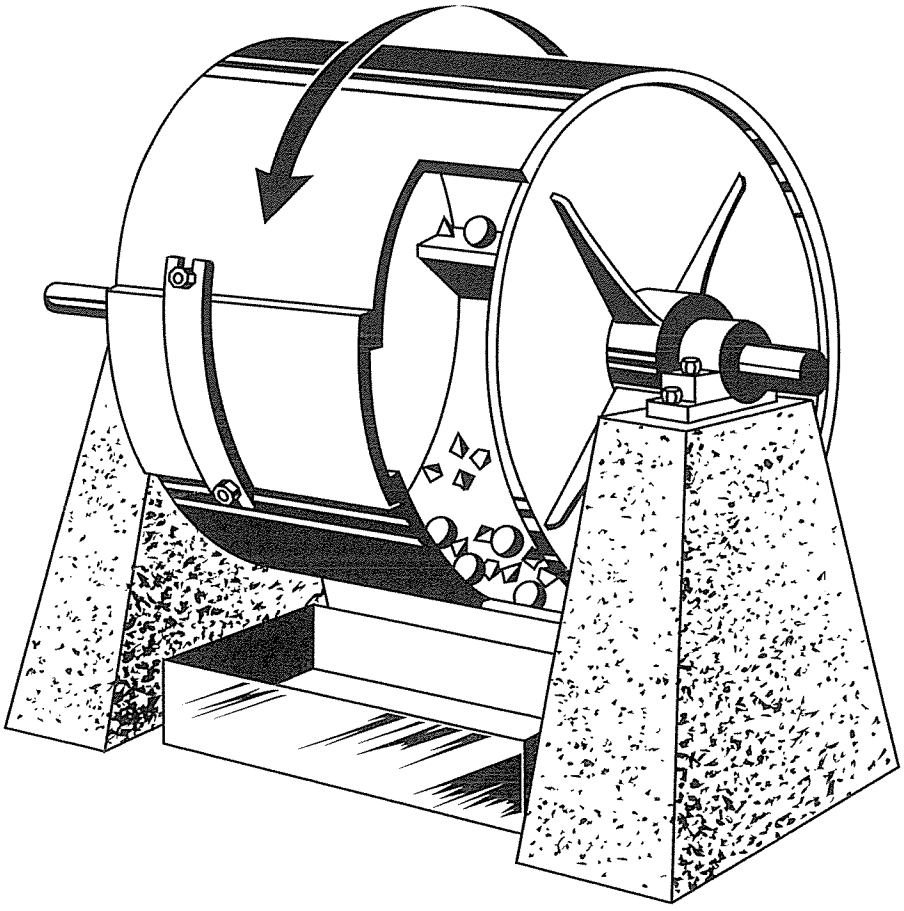


Figure 3.7 Los Angeles Abrasion Machine

aggregate is soaked in water before the test and water is used in the drum during the test. The test has been adopted by some agencies as a replacement for the Los Angeles Abrasion test.

Pore Structure, Porosity, and Absorption

The pore structure of an aggregate refers to the characteristics and configuration of voids in an aggregate particle. Pores or voids can be isolated or encapsulated within aggregate particles and they are thus impermeable, or they can be interconnected and open to the surface, making them permeable.

Porosity is the total volume of voids in the aggregate particle. Higher porosity can relate to lesser aggregate strength and lower density and specific gravity. Absorption is the intake or permeation of liquid into the pores of an aggregate particle. Higher porosity may not mean higher absorption if most of the pores are impermeable. However, high absorption does indicate a large volume of permeable voids.

An aggregate with permeable voids will absorb asphalt. Additional asphalt must be incorporated into the paving mix to compensate for the absorption of asphalt by the aggregate. This is noneffective asphalt. Such aggregates usually are not used unless less-absorptive aggregates are not available or the aggregates possess other qualities or properties that make them desirable in spite of their high-absorption rate. Blast furnace slag and many synthetic or manufactured aggregates are lightweight materials that are also highly absorptive. Their lightness and wear-resistant properties frequently outweigh the high-absorption rate consideration for pavement construction use.

AASHTO T 85 (ASTM C 127) is commonly used to determine absorption for coarse aggregates. For fine aggregates, AASHTO T 84 (ASTM C 128) is the common test standard. In these tests, absorption is expressed as the amount of water as a percentage of the total dry aggregate mass that penetrates into the pores only and does not include water that adheres to the surface of the aggregate. The following section on specific gravity provides a description of the two test procedures.

Specific Gravity

The specific gravity of an aggregate is the ratio of the weight of a unit volume of material to the weight of the same volume of water at 73°F (23°C). This property is primarily used in mix volumetric calculations for voids determination. Also, bulk specific gravities (defined in the following paragraphs) are used in the computations for adjusting quantities of the aggregate components that are to be used in an HMA mix because of the differing specific gravities of various aggregates.

The three generally accepted types of specific gravities for aggregate, depending on the defined volume of the particle, are the following:

- Apparent specific gravity
- Bulk specific gravity
- Effective specific gravity

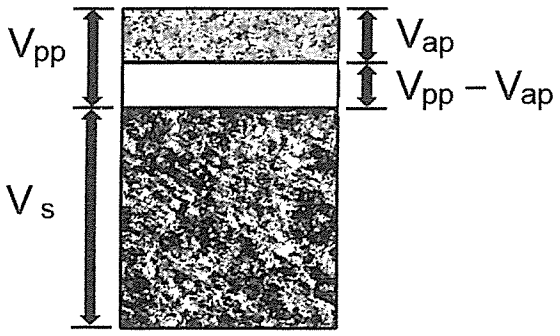


Figure 3.8 Relationship Between the Different Specific Gravities of an Aggregate Particle

As illustrated in Figure 3.8, *apparent specific gravity* considers the volume as being the volume of the aggregate itself. It does not include the volume of any pores or capillaries that become filled with water after a 24-hour soaking. *Bulk specific gravity* considers the overall volume of the aggregate particle, including the pores that become filled with water after a 24-hour soaking. The *effective specific gravity* considers the overall volume of the aggregate exclusive of the volume of pores that absorbs asphalt. Whereas bulk and apparent specific gravities can relate to individual aggregates or combined aggregates, effective specific gravity relates exclusively to the total combined aggregate structure in a mix of HMA.

Air-voids in the compacted asphalt pavement course appear throughout the mix as small pockets of air between the asphalt-coated aggregate particles. The air voids content cannot be directly measured but is back-calculated from other test results. The choice of specific gravity for the aggregate used in asphalt mix calculations can have a substantial effect on the calculated amount of air-voids in the compacted mix. The actual specific gravity of the aggregate in the mixture depends on the degree to which the aggregate absorbs asphalt. When apparent specific gravity is used, asphalt is assumed to be absorbed by all the water-permeable pores. If bulk specific gravity is used, it is assumed that the asphalt is not absorbed by the water permeable pores. Except in rare cases, neither is true. The concept of effective specific gravity more nearly approaches the correct value for determining the voids in a compacted asphalt paving mix. Bulk specific gravity may be used, however, only if an allowance for the asphalt absorbed by the aggregate is taken into account.

Equipment and procedures for determining the bulk and apparent specific gravities of coarse aggregates are outlined in AASHTO T 85 (ASTM C 127). Equipment and procedures for determining the apparent and bulk specific gravities of fine aggregates are detailed in AASHTO T 84 (ASTM C 128).

The procedure for determining the effective specific gravity of the combined aggregate in a mix actually takes place during the laboratory work for the HMA mix design. Data obtained from determining the maximum specific gravity of asphalt mixtures (ASTM D 2041) is used in calculating effective specific gravity (see Chapter 4). This important property affects the optimum binder content in a mix and the determination of the percentage of air voids in the compacted mix.

Freeze-Thaw/Weathering Resistance

Freeze-thaw resistance is an aggregate's ability to resist degradation caused by cycles of freezing and thawing when the aggregate is in a saturated condition. Internal expansive forces develop in the pores of the aggregate each time the absorbed water transforms into ice. Weathering resistance can include the freeze-thaw resistance factor, but it also refers to the aggregates' ability to withstand the volume changes and degradation that occur with cyclic wetting and drying. An aggregate's ability to resist the effects of wetting and drying cycles and freezing and thawing cycles can also be called the aggregate's durability or soundness property.

In moist and freezing climates, aggregates must exhibit a certain level of durability or soundness to resist deterioration in stockpiles in the interim stage (after aggregate processing and before HMA production). However, once the aggregates are introduced into HMA production, the need for weathering durability is debatable. Moisture within the aggregates is driven out in the dryer, and the aggregates receive a coating of impermeable asphalt binder.

The test most commonly used to determine an aggregate's durability is the sulfate soundness test—AASHTO T 104 (ASTM C 88). In this test, an aggregate sample is sieved into specified size fractions. The separated fractions are then immersed for 16 to 18 hours in a saturated solution of either sodium or magnesium sulfate for absorption of the solution into the aggregate pores. The samples are then removed from the solution, allowed to drain for a short period, and then dried in an oven until they reach constant mass. The samples are then cooled to room temperature, and the cycle of immersion and drying is repeated. Typically, five cycles are specified. At the end of the required number of cycles, each size fraction sample is washed to remove the salts, and then dried and sieved over the specified screen. The mass loss of each fraction sample is determined and expressed as a percentage loss. A weighted loss for the composite aggregate is calculated based on the original aggregate gradation. The applicability of this test to durability is given in ASTM C 88: "The internal expansive force, derived from the rehydration of the salt upon re-immersion, simulates the expansion of water on freezing."

Typical standards and agency specifications set maximum allowable loss values in the 10–20 percent range for test durations of five cycles.

Deleterious Substances

Deleterious substances are any substances in an aggregate that diminish or are detrimental to the necessary properties of HMA. Specifications for

aggregates usually contain a section in which deleterious materials are named, and a limit is placed on the amount of each that is allowable in the aggregate. Typically, identified deleterious materials include the following:

- Clay and other surface coatings on aggregate particles
- Clay lumps
- Soft or weak particles
- Organic materials

Aggregate cleanliness is one way of describing the level of surface coatings on an aggregate. Aggregate coatings can affect the bond or adhesion of binder to the aggregate. Aggregate cleanliness of coarse aggregate is often determined by visual inspection, but a washed sieve analysis generally provides positive proof. The sand-equivalent test, developed by the California Division of Highways and described in AASHTO T 176 (ASTM D 2419), is the most common method of assessing the level of cleanliness of fine aggregates by determining the relative proportion of detrimental fine dust or claylike materials. The test method summary describes the procedure: "A measured volume of soil or fine aggregate and a small quantity of flocculating solution are poured into a graduated plastic cylinder and are agitated to loosen the claylike coatings from the sand particles in the test specimen. The specimen is then 'irrigated' using additional flocculating solution forcing the claylike material into suspension above the sand. After a prescribed sedimentation period, the height of flocculated clay is read and the height of sand in the cylinder is determined. The sand equivalent is the ratio of the height of sand to the height of clay times 100." Cleaner aggregates will have higher sand equivalent values. Common specification minimum limits for this property range from 40 to 50.

Determination of liquid limit and plasticity index properties is another method sometimes required by specifications for assessing the clay content in fine aggregate. Procedures for determining these properties are contained in AASHTO T 89 and T 90 (ASTM D 4318). These procedures define *liquid limit* as "the water content, in percent, of a soil at the arbitrarily defined boundary between the semi-liquid and plastic states." *Plasticity index* is defined as "the range of water content over which a soil behaves plastically."

Samples for both the liquid limit and the plastic limit tests use the portion of the material that passes the No. 40 (0.425 millimeter) sieve mixed with varying amounts of water. In the liquid limit procedure, a prepared sample is spread in a brass cup and cut with a grooving tool, forming a notch through the sample. The brass cup is then repetitively raised and dropped until the divided sample sections flow together. The liquid limit is the corresponding water content at which the sample flowed together at 25 drops of the cup.

In order to determine the plasticity index, the plastic limit of the material must be determined. According to the test method summary of D 4318, "the plastic limit is determined by alternately pressing together and rolling into a ⅛-inch (3.2 millimeters) diameter thread a small portion of plastic soil until its water content is reduced to a point at which the thread crumbles and can no longer be pressed together and re-rolled. The water content of the soil at this point is reported as the plastic limit. . . . The plasticity index is calculated as the difference between the liquid limit and the plastic limit."

Soft and weak particles are often identified using AASHTO T 112 (ASTM C 142). Such particles often break down in stockpiles when they are exposed to moisture, thus creating smaller particles and unwanted fines. In this procedure, the particles finer than the No. 200 (0.075 millimeter) sieve are removed by washing the material following the procedures detailed in AASHTO T 11 (ASTM C 117). The material is then separated into fractions of designated sizes and soaked in water for 24 hours. The soaked particles are each individually squeezed between fingers to break down into smaller pieces any material that might be clay lumps or friable particles. The material is then washed over a designated sieve, and the dry mass of the retained material is determined. The amount of clay lumps and friable particles is calculated by determining the amount of material passing the designated wash screen and expressing it as a percentage of the original sample mass before the 24-hour soak.

For HMA, **organic materials** are usually identified on a visual basis and specifications typically reflect this.

Examples of typical aggregate quality requirements are shown in Tables 3.4 and 3.5.

Examples of Common HMA Aggregate Specifications

Table 3.4
AASHTO M 323, Superpave Aggregate Consensus Properties

ESALs (mil)	Fractured Faces Coarse Aggregate, % min.		Uncompacted Voids Content of Fine Aggregate, % min.		Sand Equivalent % min.	Flat and Elongated, % max.
	Depth from Surface		Depth from Surface			
	≤ 100 mm	> 100 mm	≤ 100 mm	> 100 mm		
< 0.3	55/–	–/–	–	–	40	–
0.3 to < 3	75/–	50/–	40	40	40	10
3 to < 10	85/80	60/–	45	40	45	10
10 to < 30	95/90	80/75	45	40	45	10
≥ 30	100/100	100/100	45	40	50	10

Table 3.5
AASHTO MP 8, Stone Matrix Asphalt (SMA) Aggregate Quality Requirements

Coarse Aggregate		Fine Aggregate	
Los Angeles Abrasion, %	30 max.		
Flat & Elongated, %		Soundness (5 cycles), %	
3:1	20 max.	Sodium Sulfate	15 max.
5:1	5 max.	Magnesium Sulfate	20 max.
Absorption, %	2.0 max.		
Soundness (5 cycles), %			
Sodium Sulfate	15 max.	Liquid Limit, %	25 max.
Magnesium Sulfate	20 max.		
Crushed Content			
One Face	100 min.	Plasticity Index	Nonplastic
Two Faces	90 min.		

3.2 Composite Aggregate Analysis

3.2.1 Introduction

HMA is normally composed of a blend of individual aggregates that make up the composite aggregate structure, or matrix, of the mix. These individual aggregates are stockpiled separately, are typically different in size and gradation, and might even come from different sources. Particularly during the processing stages, it is important that the individual aggregate components possess certain qualities. It is also important that the composite of the blended aggregates possess key characteristics critical to the HMA. These qualities and characteristics are reflected in most specifications that stipulate requirements for both the composite aggregates and the individual aggregate components. One of the key characteristics of the composite aggregate is gradation.

3.2.2 Gradation

Gradation Types

Composite aggregates can fall into one of the following types of gradation or categories of particle size distributions as illustrated in Figure 3.9:

- Well-graded or dense-graded
  - o Coarse-graded
  - o Fine-graded
- Open-graded

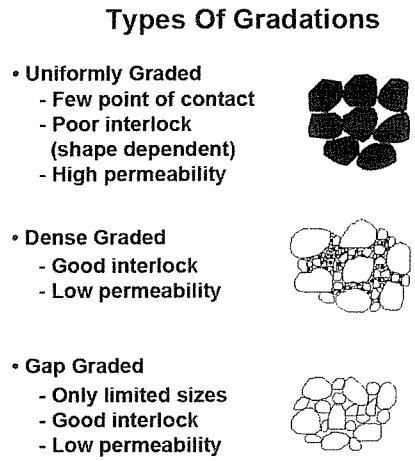


Figure 3.9 Gradation Types

- Uniformly graded (a single-size aggregate)
- Gap-graded

A uniformly graded aggregate structure consists of particles that are primarily of one size or their range of sizes is very narrow. The amount of particle-to-particle contact is low, and the interlock of the aggregates is low. The amount of voids between the particles is high; thus, permeability of the aggregate structure is high. Uniformly graded aggregates are not usually desired in asphalt mixtures.

Some asphalt mixes have an aggregate structure that is called open-graded. This is actually a uniformly graded aggregate structure with a very small amount of fine-aggregate particles. Open-graded mixes are most often used for their characteristics of high permeability and high surface friction. Surface courses made with these mixes are also called "porous friction courses."

A well-graded aggregate structure has an extensive distribution of particle sizes ranging from fine to coarse. The smaller aggregate particles fill in gaps and voids between larger particles. The grading is considered dense in that the concentration of aggregate particles is high and the amount of voids between aggregate particles is small. Permeability of the aggregate structure is low, and interlock between aggregate particles is high. Paving mixtures with such aggregate structures may be easily compacted but tend to tightly pack and have air-void contents that are too low.

Gap-graded aggregates refer to the gradation characteristic in which there is a "gap" in the particle-size distribution. There is a range of particle size fractions (typically the mid-size range) that is either devoid of particles or the amount of particles in that range is low. Gap-graded aggregates are not as dense in particle concentration as well-graded aggregates. In turn, the amount of voids between aggregate particles and the permeability are higher than that of well-graded aggregates. Many times, asphalt mixtures must necessarily have the greater voids content of a gap-graded aggregates so that the mix can contain an adequate amount and coating thickness of asphalt binder. Gap-graded aggregate structures typically have good interlock between particles.

Two other dense-gradation types that are often mentioned are "coarse-graded" or "fine-graded." Fine-graded aggregate structures usually contain a greater amount of fine aggregate, generally resulting in HMA mixtures with higher asphalt content, a lesser compactive effort needed during placement, a more closed-surface texture, and an increased potential for surface deformation. Coarse-graded aggregate structures, on the other hand, contain a lesser amount of fine aggregate. HMA mixtures with a coarse grading tend to have a lower asphalt content, require more compactive effort, have a more open-surface texture, and have a lower potential for surface deformation.

Composite-Aggregate Size Designations

As discussed in the previous section, the various types of aggregate gradations each create different characteristics in asphalt mixtures and each can serve a different purpose. In addition to grading type, the maximum size of the aggregate is critical in paving mixtures because it also affects key

properties and dictates the thickness at which a paving mat can practically be placed. Typically, the lift thickness should be at least 2 to 2½ times the maximum aggregate size in the mix. In general terms, larger aggregates in mixes usually create better stability and skid resistance in the pavement but are prone to segregation and lower workability during construction. Conversely, mixes with smaller maximum-size aggregates are usually more workable and less prone to segregation, and contribute to better surface smoothness.

In specifications for asphalt mixtures, the gradation requirements for the aggregate structure need to be established for the appropriate maximum (and the nominal maximum) size of aggregate and the type of aggregate gradation. Maximum and nominal maximum sizes set the upper end of the grading band, and the grading type influences the band or range of maximum and minimum values for each identified sieve size. Examples of common aggregate sizes and their corresponding gradation specifications are contained in Table 3.6.

Composite-Aggregate Size Fractions

The composite-aggregate gradation is usually considered to have three major fractions: coarse aggregate, fine aggregate, and mineral filler. For many years, the dividing size between coarse and fine aggregate was the No. 8 (2.36 millimeters) sieve, but many specifications today stipulate the No. 4 (4.75 millimeters) sieve as the dividing size. The separation between fine aggregate and mineral filler has been and remains at the No. 200 (0.075 millimeter) sieve. Thus, the fractions are broken down this way:

- Coarse aggregate is the portion of material retained on the No. 4 (4.75 millimeters) sieve.
- Fine aggregate is the portion of material passing the No. 4 (4.75 millimeters) sieve and retained on the No. 200 (0.075 millimeter) sieve.
- Mineral filler is the portion passing the No. 200 (0.075 millimeter) sieve.

0.45 Power Gradation Curve

Grading curves are very helpful in assessing aggregate-gradation characteristics and in making necessary adjustments in mix designs. Since its development in the 1960s, the Federal Highway Administration grading chart is the most widely accepted method for plotting gradations for comparison to the maximum density grading. The chart is based on the Fuller maximum density equation of

$$p = 100\left(\frac{d}{D}\right)^{0.5}$$

In the development of the FHWA chart, however, the equation exponent of 0.45 was used in lieu of the 0.5. The actual equation for the FHWA chart is

$$p = 100\left(\frac{d}{D}\right)^{0.45}$$

in which

p = total percentage passing given sieve
d = size of sieve opening
D = largest size (sieve opening) in gradation

Table 3.6
Typical Mix Grading Sizes and Compositions (from ASTM D 3515)

Sieve Size	Mix Designations and Nominal Maximum Size of Aggregate				
	1½ in (37.5 mm)	1 in (25.0 mm)	¾ in (19.0 mm)	½ in (12.5 mm)	⅜ in (9.5 mm)
	Total Percent Passing (by weight)				
2 in (50 mm)	100	–	–	–	–
1½ in (37.5 mm)	90–100	100	–	–	–
1 in (25.0 mm)	–	90–100	100	–	–
¾ in (19.0 mm)	56–80	–	90–100	100	–
½ in (12.5 mm)	–	56–80	–	90–100	100
⅜ in (9.5 mm)	–	–	56–80	–	90–100
No. 4 (4.75 mm)	23–53	29–59	35–65	44–74	55–85
No. 8 (2.36 mm)*	15–41	19–45	23–49	28–58	32–67
No. 16 (1.18 mm)	–	–	–	–	–
No. 30 (0.60 mm)	–	–	–	–	–
No. 50 (0.30 mm)	4–16	5–17	5–19	5–21	7–23
No. 100 (0.15 mm)	–	–	–	–	–
No. 200 (0.075 mm)**	0–6	1–7	2–8	2–10	2–10
Bitumen, Weight Percent of Total Mixture***	3–8	3–9	4–10	4–11	5–12
	Suggested Coarse-Aggregate Size Numbers				
	4 and 67 or 4 and 68	5 and 7 or 57	67 or 68 or 6 and 8	7 or 78	8

* In considering the total grading characteristics of a bituminous paving mixture, the amount passing the No. 8 (2.36 mm) sieve is a significant and convenient field control point between fine and coarse aggregate. Grading approaching the maximum amount permitted to pass the No. 8 (2.36 mm) sieve will result in pavement surfaces having comparatively fine texture, while gradings approaching the minimum amount passing the No. 8 (2.36 mm) sieve will result in surfaces with comparatively coarse texture.

** The material passing the No. 200 (75 μm) sieve may consist of fine particles of the aggregates or mineral filler, or both, but shall be free of organic matter and clay particles. The blend of aggregates and filler, when tested in accordance with Test Method D 4318, shall have a plasticity index of not greater than 4, except that this plasticity requirement shall not apply when the filler material is hydrated lime or hydraulic cement.

*** The quantity of bitumen is given in terms of weight percentage of the total mixture. The wide difference in the specific gravity of various aggregates, as well as a considerable difference in absorption, results in a comparatively wide range in the limiting amount of bitumen specified. The amount of bitumen required for a given mixture should be determined by appropriate laboratory testing or on the basis of past experience with similar mixtures, or by a combination of both.

In the FHWA chart (see Figure 3.10), values for cumulative percent passing are plotted on the vertical axis and the particle or sieve sizes (in millimeters) raised to the 0.45 power are plotted on the horizontal axis. The maximum-density grading is represented by a straight line from the origin at the lower left of the chart to the intersection of the desired maximum particle size

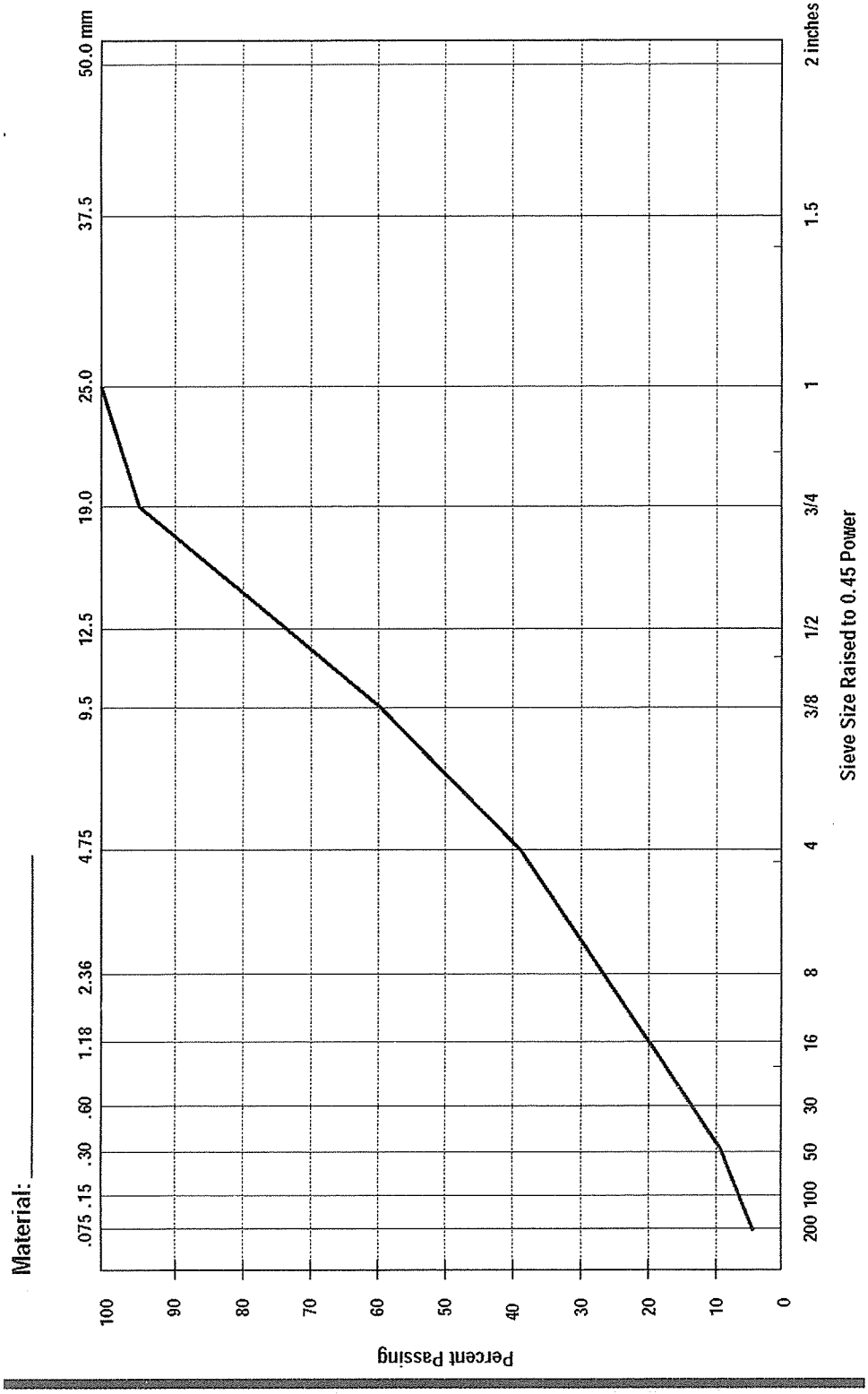


Figure 3.10 FHWA Chart

and the 100 percent passing line at the upper right of the chart. This line is referred to as the maximum-density line. (See "Maximum Particle Size and Grading" in this chapter for the definition of maximum particle size.)

Dense-graded (or well-graded) aggregate structures plot near and along the maximum-density line. Mixes with such gradings tend to have low Voids in Mineral Aggregate (VMA). Plots of gap-graded and open-graded aggregates gradation curve out or away from the maximum-density line (above or below). Greater VMA is achieved with these aggregate structures. Those gradations that generally curve below the maximum-density line are considered coarse gradings, and those that generally curve above the line are considered fine gradings. See Figure 3.11 for examples of plotted gradations.

Gradation Blending Calculations

Combining two or more aggregates having different gradations to produce an aggregate blend that meets grading specifications (or even a specific desired grading) for a particular asphalt mix is common in asphalt construction. A number of methods are available for determining relative amounts of various aggregates needed to obtain a desired gradation.

Regardless of the number of aggregates combined or of the method by which the proportions are determined, the basic formula expressing the combination is

$$P = Aa + Bb + Cc + \ldots \quad \text{(Equation 1)}$$

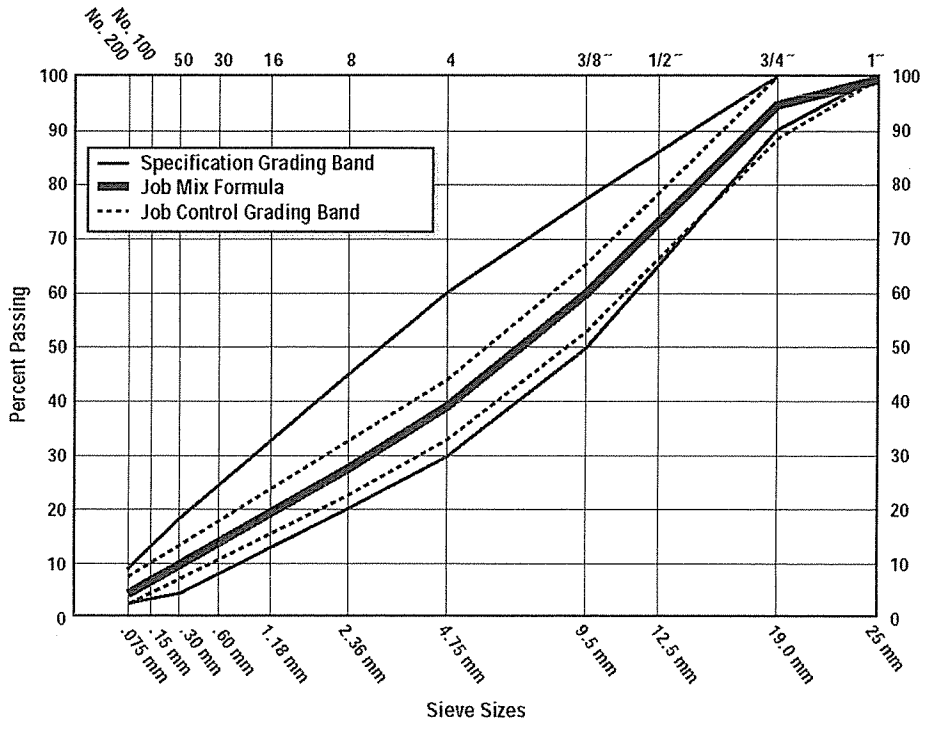


Figure 3.11 Example Gradation Curves

where

P = the percentage of material passing a given sieve for the combined aggregates

A, B, C = percentages of material passing a given sieve for the individual aggregates A, B, C, etc.

a, b, c = proportions of individual aggregates, A, B, C, etc., used in the combination and where the total = 1.00

The combined percentages (P, in equation 1) for each sieve size should closely agree with the percentages of the desired composite aggregate grading. None of the percentages should fall outside the specified grading limits. Obviously, there may be several acceptable combinations. An optimum combination is achieved when the blended or composite percentages match as closely as possible to the original desired percentages. If the blended gradation falls outside the specifications, the responsible proportion in the blend should be modified, with the other proportions also modified to total 100 percent. Sometimes, it is necessary to delete a proposed aggregate and obtain replacement material if a satisfactory blend cannot be developed with the aggregate initially suggested.

Mathematical methods have been developed that will directly determine an "optimum" combination of aggregates. To accomplish this calculation, various optimization techniques have been used, such as the closest to the middle of the specification range. The speed of these methods varies with the number of stockpiles being blended. Many designers use computer spreadsheet programs to quickly evaluate a number of alternative blends. The Asphalt Institute's SW-2, *Mix Design Program* computer program can also be used to visually evaluate the gradation plot of numerous blends very rapidly. Regardless of which method is used, a trial-and-error approach guided by reasoning, is usually the easiest and best procedure to determine and refine a satisfactory combination of aggregates. A computer-generated blend should not be used without further evaluation and checking.

Table 3.7 is an example of a simple computer spreadsheet in which composite gradations can be calculated. Equation 1 has been entered into the pertinent cells as a "formula" so that a composite gradation is instantly calculated when trial proportion values and individual aggregate gradations are entered into their respective cells. Numerous trial blends can be evaluated in a short time.

When only two or three aggregates are blended for a composite, there are some simple techniques for evaluating the needed proportions of the components. For combining two aggregates, the basic formula of Equation 1 condenses to the following:

$$P = Aa + Bb \quad \text{(Equation 2)}$$

Since $a + b = 1$, then $a = 1 - b$.

Substituting this into Equation 2, and solving for b:

$$b = \frac{P - A}{B - A} \quad \text{(Equation 3)}$$

Table 3.7
Example Blending Table

Aggregate:		Component Materials						
		RAP	No. 56 Coarse Agg.	No. 7 Intermed. Agg.	Crusher Fines	Natural Sand	Total Percentage	
Percent Used:		13	18	22	37	10	100	¾ in Grading Band
Sieve		Individual Gradations					Composite	
1 in	25 mm	100	100	100	100	100	100	100
¾ in	19 mm	100	68	100	100	100	94	90–100
½ in	12.5 mm	95	16	95	100	100	83	
⅜ in	9.5 mm	88	4	50	100	100	70	56–80
No. 4	4.75 mm	71	2	4	83	94	51	35–65
No. 8	2.36 mm	58	2	2	59	86	39	23–49
No. 16	1.18 mm	48	2	1	42	78	30	
No. 30	0.6 mm	39	1	1	33	58	23	
No. 50	0.3 mm	26	1	1	23	24	15	5–19
No. 100	0.15 mm	14	1	1	15	9	9	
No. 200	0.075 mm	7	1	1.2	10.9	5.1	5.9	2–8

An expression for a can also be found:

$$a = \frac{P - B}{A - B} \quad \text{(Equation 4)}$$

Assume that a single aggregate stockpile is to be blended with sand to meet grading requirements for an asphalt paving mixture. These are given in Table 3.8a as aggregates *A* and *B*. To make a determination of acceptable proportions of the two materials:

1. Examine the two gradations to determine which aggregate will contribute most of the material for certain sizes. In this case, most of the minus No. 8 (2.36 millimeters) aggregate will be furnished by aggregate *B*.
2. Using the percentages for the No. 8 (2.36 millimeters) sieve and substituting into Equation 3, the proportions are determined to meet the midpoint of the specification (Table 3.8b).
3. Inspection of the first trial gradation shows the percent passing the No. 200 (0.075 millimeter) sieve to be close to the lower specification limit. Increase the proportion of aggregate *B* (in this case to 0.55) and compute the gradation of the second trial (Table 3.8c).
4. Inspection now shows the gradation is critical on the No. 30 (0.600 millimeter) sieve. Reduce the proportion of aggregate *B* to 0.52 and compute the gradation of the third trial (Table 3.8d). The third trial is a "best fit" considering the percents passing the No. 30 (0.600 millimeter) and No. 200 (0.075 millimeter) sieves.